Combinations

1 Evaluate the following expressions:

a $5!$ b $4!6!$ c $\frac{11!}{5!6!}$ d $\frac{19!}{15!4!}$

e $\frac{5!}{4!}$ f $\frac{10!}{5!4!}$ g $\binom{7}{2}$ h ${}^{4}C_0$

i $\binom{5}{5}$ j $\binom{7}{3}$ k $\binom{10}{9}$ l $\frac{{}^{10}C_5}{{}^{10}C_6}$

2 Simplify:

a $n\,(n-1)!\,(n+1)$ b $\frac{(n+1)!}{n!}$

c $\frac{n!}{(n-2)!}$ d $\frac{(n+4)!}{(n+6)!}$

3 Find the value of n if:

a $\binom{n}{3} + \binom{n}{2} = 8\binom{n}{1}$

b The ratio of the number of combinations of $2n+2$ different objects taken n at a time to the number of combinations of $2n-2$ different objects taken n at a time is $14:1$.

4 Prove the following:

a $\binom{n}{0} = 1$ b $\binom{n}{1} = n$

c $\binom{n}{0} = {}^{n}C_n$ d $\binom{100}{100} = 1$

e $\binom{100}{100-1} = 100.$ f $\binom{100}{100-40} = \binom{100}{40}$

Applications

5 Ben has 3 shirts, each in a different colour: crimson (C), pink (P) and white (W), and 4 ties, each in a different colour: blue (B), grey (G), red (R) and yellow (Y).

a Construct a table to show all the possible combinations Ben could wear.

b How many different combinations are possible?

6 A hotel is supervised by a team of 6 security guards at any given time during the day. If there are 12 security guards available, how many different ways can a team of guards be chosen?

7 A boss wants to select one group of 4 people from his 28 staff. How many different groups

8 A group of 28 students visit the tenpin bowling centre and must organise themselves into a group of 8 to use a lane. How many different ways are there for a group of 8 students to be formed?

9 A boss wants to select one group of 3 people from his 27 staff. How many different groups are possible?

10 A restaurant offers ten different sauces of spaghetti. Grace chooses two sauces for a spaghetti. How many different possible choices could she make?

11 A variety pack of candy consists of nine bars each with a different flavour. If four bars of candy are chosen at random, how many different selections are possible?

12 In a Lotto draw, 5 different numbers are chosen from the numbers 1 to 24. How many possible selections are there? Assume that order does not matter.

13 A variety pack of chocolate consists of seven bars, each with a different flavour. If three bars of chocolate are chosen at random, how many different selections are possible?

14 Quiana is ordering a salad at a restaurant. She can order a salad consisting of just lettuce, or she can also add any of the following ingredients:

$$\text{Onions, Cucumbers, Tomatos, Cheese, Olives, Carrots, Mushrooms}$$

How many different variations of a salad can she possibly choose?

15 Out of the 16 cricket players, only 12 are selected for the team.

a How many different teams can be made?

b Two opening batsmen are to be selected from among the 12 chosen players. How many combinations of opening batsmen are possible?

16 In court, a jury panel of 8 members is to be made up from a group of 20 candidates. In how many different ways can a panel be formed?

17 In a bicycle race, a quinella is a bet on the first 2 bicycles that finish the race, but the order in which these 2 bicycles place does not matter. How many different quinella bets are possible for a bicycle race where 14 bicycles are competing?

18 In a game of football, each team must have 1 goalkeeper and 10 other players on the field. A football squad consists of 17 players, 2 of which are goalkeepers. In how many ways can players be chosen to start the game?

19 A tea shop sells 10 different types of tea a[illegible]different sets of teacups, and has room to display 6 types of tea and 3 sets of teacups. The owner changes the display each month. How many different display combinations of tea and tea cups are possible?

20 A teacher wants to divide her 12 students into a group of 4, then a group of 5 and finally a group of 3. In how many ways can she do this?

21 Determine the number of ways can 3 cards be chosen from a standard deck of 52 cards if:

- **a** Exactly one of the cards is to be a Queen.
- **b** None of the cards is to be a Queen.
- **c** At most one of the cards is to be a Queen.

22 Amelia is playing a game of cards. She will randomly draw 5 cards from a standard deck of 52 cards, and wants only 3 of the cards to have clubs on them. How many different selections would contain the cards she wants?

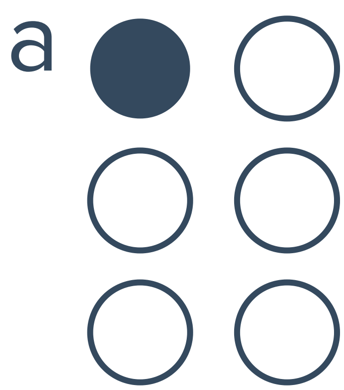
23 Determine the number of ways can 6 people be chosen from a group of 11 people if:

- **a** The eldest person is included.
- **b** The eldest person is not included.
- **c** The eldest and youngest people are both to be included.

24 How many different unordered selections can be made from a set of 7 elements?

25 How many different unordered selections can be made from the set of letters of the Greek alphabet, which contains 24 letters?

26 In the Braille system of writing, six dots are arranged, and a particular combination of raised dots represents a particular letter. The letter 'a' is represented by one dot being raised, as shown:



- **a** In how many ways can two dots be raised?
- **b** In how many ways can 1 to 6 dots be raised?

27 A selection of 4 people are to be chosen from a group of 7 people. How many selections are possible if the youngest or oldest is included but not both?

28 A team of 3 is to be chosen at random from a group of 5 girls and 6 boys. Determine the number of ways can the team be chosen if:

a There are no restrictions.

b There must be more boys than girls.

29 Frank's organisation wants to form a partnership with a charity. After a day of speaking with representatives from different charities, he looks in his wallet and sees that he has 3 identical business cards from one charity and one business card from each of the other 9 charities he spoke with. If he wants to select 3 cards to keep, no two from the same charity, how many different selections can he make?

30 The state government needs to determine how school funding is allocated for the following year. Of the 8 public schools that will be affected, 5 of them are asked to attend a meeting. Determine the number of ways the 5 representative schools can be chosen if:

a The two least funded schools must attend.

b Neither of the two least funded schools can attend.

c Saint Hibiscus Girls' and Saint Hibiscus Boys' are two of the public schools affected, and exactly one of them must be chosen.

31 5 boys and 6 girls are part of the debate team. 4 of them must represent the school at the upcoming debate tournament. Determine the number of ways the team of 4 can be formed if:

a It must contain 2 boys and 2 girls.

b It must contain at least 1 boy and 1 girl.

32 A company has recently expanded overseas and is looking to recruit 5 new Resource Analysts. They have narrowed it down to 8 equally qualified applicants, of which 2 of them can speak a second language. In how many ways can they fill the positions if exactly one of the recruits must speak a second language?

33 In a court case, the defendant can choose a jury of 9 members from 15 candidates consisting of 9 women and 6 men. They notice that one particular lady looks ill and wish to avoid choosing her. How many different panels of jury members can they create if they want at least 7 women to make up the jury?

34 A diagonal of a polygon is a line joining two non-adjacent vertices.

a How many diagonals does a 5-sided polygon have?

b How many diagonals does a 6-sided polygon have?

c How many diagonals does a n-sided polygon have?

35 Yuri constructs 13 straight lines, no two of which are parallel. Three of the lines are concurrent (intersect at a single point). How many points of intersection are there?

36 In a radio discussion about an upcoming election, 7 Labour voters, 5 Liberal voters and 5 Greens voters called up to have a say. The radio announcers want to choose 1 Labour voter, at least 4 Liberal voters and at least 3 Greens voters to talk to on air. In how many ways can they select the callers?

37 A senior science teacher wants to receive feedback from a selection of students on whether they thought the exam was fair. The table shows the range of marks and the number of students who fell within each range:

Score range	$0-40$	$41-50$	$51-60$	$61-70$	$71-80$	$81-100$
Number of students	2	4	5	7	10	5

She selects 2 students to provide feedback, but does not want more than 1 student from any score range. In how many ways can she select the 2 students?

38 10 red balls and 12 green balls are in a bag. 3 balls are to be selected from the bag at random, without replacement.

a Find the total number of ways that the 3 balls can be chosen.

b Find the total number of ways that 3 red balls can be chosen.

c Hence, find the probability that 3 red balls are chosen.

39 A company consists of 17 males and 12 females. 6 employees are selected at random without replacement.

a Find the total number of ways that the 6 employees can be selected.

b Find the total number of ways that 6 male employees can be selected.

c Hence, find the probability that 6 male employees are selected.

40 A class of 24 students has 19 people whose birthday is in April. 4 students are selected at random without replacement.

a Find the total number of ways that the 4 students can be selected.

b Find the total number of ways that 4 students can be selected such that none of their birthdays are in April.

c Hence, find the probability that the 4 students are selected do not have their birthdays in April.